

Exp No: 27

BINARY SEMAPHORE IN SYSTEMC

PROGRAM:

```
#include <systemc.h>
```

```
SC_MODULE(binary_semaphore) {
```

```
    int sem;
```

```
    sc_event ev;
```

```
    void task1() {
```

```
        while (true) {
```

```
            wait(1, SC_SEC);
```

```
            if (sem == 1) {
```

```
                sem = 0;
```

```
                cout << "Task1 Entered Critical Section at "
```

```
                    << sc_time_stamp() << endl;
```

```
                wait(2, SC_SEC);
```

```
                sem = 1;
```

```
                ev.notify();
```

```
            }
```

```
        else {
```

```
            wait(ev);
```

```
        }
```

```
    }
```

```
}
```

```

void task2() {
    while (true) {
        wait(2, SC_SEC);

        if (sem == 1) {
            sem = 0;

            cout << "Task2 Entered Critical Section at "
                << sc_time_stamp() << endl;

            wait(1, SC_SEC);

            sem = 1;
            ev.notify();
        }
        else {
            wait(ev);
        }
    }
}

SC_CTOR(binary_semaphore) {
    sem = 1;

    SC_THREAD(task1);
    SC_THREAD(task2);
}

};

int sc_main(int argc, char* argv[]) {

    binary_semaphore obj("Binary_Semaphore");

```

```
sc_start(10, SC_SEC);
```

```
return 0;
```

```
}
```

OUTPUT

The screenshot shows the EDA Playground web interface. The browser address bar displays `https://www.edaplayground.com/x/5Vg4`. The interface includes a sidebar with navigation links for Languages & Libraries, Tools & Simulators, and Examples. The main editor area contains two files: `testbench.cpp` and `design.cpp`. The `testbench.cpp` file contains the following code:

```
46     }  
47     }  
48     }  
49     }  
50     SC_CTOR(binary_semaphore) {  
51         sem = 1;  
52         SC_THREAD(task1);  
53         SC_THREAD(task2);  
54     };  
55     };  
56     };  
57     };  
58     int sc_main(int argc, char* argv[]) {  
59         binary_semaphore obj("Binary_Semaphore");  
60         sc_start(10, SC_SEC);  
61         return 0;  
62     }  
63     }  
64     }
```

The `design.cpp` file contains the following code:

```
1 // Code your design here.  
2 // Uncomment the next line for SystemC modules.  
3 // #include <systemc>
```

The output window shows the execution results:

```
[2026-08-25 04:04:17 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim  
Compile done. Starting run...  
  
SystemC 2.3.3-Accellera --- May 23 2020 14:33:56  
Copyright (c) 1996-2018 by all Contributors,  
ALL RIGHTS RESERVED  
Task1 Entered Critical Section at 1 s  
Task1 Entered Critical Section at 4 s  
Task1 Entered Critical Section at 7 s  
Done
```